

Graph Surgery and the Do-Operator

A Precise Correspondence for Acyclic Structural Causal Models

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Abstract. The do-operator is described graphically by deleting arrows into its targets and functionally by replacing their mechanisms with constants. To call these operations equivalent is not yet a mathematical statement: one returns a graph and remembers only the targets, whereas the other returns mechanisms and also remembers the imposed values. We make a dependency-level comparison precise for deterministic acyclic structural causal models with finitely many endogenous variables. If $\text{Graph}(F)$ extracts the dependencies of a mechanism family F , our main theorem is

$$\text{Graph}(F') = \text{Surg}(\text{Graph}(F), T_t).$$

Thus replacing target mechanisms removes exactly the dependencies removed by graph surgery. For a model $M = (G, F)$ whose graph may contain unused arrows, we characterize when the same equality holds with G in place of $\text{Graph}(F)$; it holds for every intervention exactly when G records the dependencies of F exactly. We then define the intervened model, characterize its run, show how sequential interventions combine, and prove that an outcome depends only on interventions at its actual dependency ancestors.

Keywords: Structural causal models · Interventions · Do-operator · Graph surgery · Dependency graphs.

1 Introduction

The do-operator is the standard notation for an intervention in a structural causal model [4]. An expression such as $\text{do}(A = a)$ says that A is no longer computed by its ordinary mechanism: it is supplied by the experimenter and held at a . A formal semantics should identify the resulting model and explain how it behaves.

Zhang conjectured that deleting incoming arrows and replacing mechanisms by constants are equivalent views of intervention [6, Sec. 6.4]. The word “equivalent” needs to be made precise. Graph surgery and mechanism replacement are not literally the same operation: they act on different objects and, more importantly, graph surgery cannot distinguish $\text{do}(A = 0)$ from $\text{do}(A = 1)$. A

natural comparison is between the dependencies that remain. We formulate that comparison and prove it for deterministic acyclic structural causal models with finitely many endogenous variables.

Our starting point is a causal model $M := (G, F)$ with two components. The directed acyclic graph G describes which variables may directly affect which others, and the mechanism family F says how their values are computed. Compatibility requires every dependency used by F to occur as an arrow of G ; it does not require every arrow to be used. We call the model exact when the arrows of G are precisely the dependencies of F . For an exogenous state u , $\text{Run}(M, u)$ is the unique world generated by the mechanisms.

We represent a simultaneous intervention by a type-respecting partial map ι on V , whose domain T_ι is its target set. On the graph, $\text{Surg}(G, T_\iota)$ deletes the arrows entering the targets. In the equations, F^ι replaces the corresponding mechanisms with the constants supplied by ι . Our central result says that the same dependency graph is reached in either order: update the mechanisms and then extract their dependencies, or first extract the dependencies and then perform graph surgery. When the supplied graph G is exact, this is also the graph obtained by performing surgery directly on G .

Our main results are:

1. an exact correspondence between graph surgery and constant mechanism replacement (Theorem 1);
2. a characterization of when this correspondence agrees with the graph supplied as part of a causal model (Corollary 1);
3. a law for combining sequential interventions, including the fact that a later assignment replaces an earlier one (Theorem 2); and
4. an ancestor theorem: the value of a set of variables depends only on the part of the intervention applied to their actual dependency ancestors (Theorem 3).

2 Structural causal models

Let V be a finite set of endogenous variables. Each $v \in V$ has a nonempty value set $\mathcal{X}_v$. Let

$$\mathcal{X} := \prod_{v \in V} \mathcal{X}_v$$

be the set of complete assignments, which we call *worlds*. For $S \subseteq V$, write x_S for the restriction of a world $x \in \mathcal{X}$ to S . Let $\mathcal{U}$ be a nonempty set of exogenous states. An element $u \in \mathcal{U}$ may collect local disturbances, shared background factors, or any other information fixed outside the endogenous equations. No probability law is needed.

A mechanism family on V is a collection of functions

$$F_v : \mathcal{U} \times \mathcal{X} \longrightarrow \mathcal{X}_v \quad (v \in V).$$

We give every mechanism the common domain $\mathcal{U} \times \mathcal{X}$ deliberately. This avoids building an a priori parent set into the type of F_v and lets the dependencies be

recovered from the functions themselves: a coordinate that F_v ignores contributes no arrow. Compatibility will require every coordinate that can affect F_v to be permitted by G , while acyclicity will make the resulting equations evaluable in topological order.

Definition 1 (Dependency graph). *A mechanism F_v depends on $w \in V$ if changing only w can change the value returned by F_v . That is, there are $u \in \mathcal{U}$ and worlds $x, y \in \mathcal{X}$ that agree at every variable except possibly w , but for which $F_v(u, x) \neq F_v(u, y)$. The dependency graph of F , written $\text{Graph}(F)$, has vertex set V and an arrow $w \rightarrow v$ exactly when F_v depends on w .*

Thus the graph records actual rather than merely permitted dependencies.

Definition 2 (Causal model). *A deterministic acyclic structural causal model is a pair $M := (G, F)$, where G is a directed acyclic graph and F is a mechanism family compatible with it: every arrow of $\text{Graph}(F)$ is also an arrow of G . A variable w is a parent of v when G has an arrow $w \rightarrow v$. The model is exact when $G = \text{Graph}(F)$.*

Compatibility gives the graph and mechanisms distinct roles. The graph says which dependencies are permitted; the mechanisms determine which of them are used in a particular model. For example, G may contain $w \rightarrow v$ even when F_v ignores w ; such a model is compatible but not exact. Although F_v is written with a whole world as input, compatibility ensures that its value depends only on the parents of v .

Lemma 1 (Only parents matter). *If $M = (G, F)$ is a causal model and two worlds $x, y \in \mathcal{X}$ agree on every parent of v , then*

$$F_v(u, x) = F_v(u, y)$$

for every $u \in \mathcal{U}$.

Proof. The set V is finite. Pass from x to y by changing one variable at a time. Only nonparents need to be changed. If w is not a parent of v , compatibility implies that $w \rightarrow v$ is not an arrow of $\text{Graph}(F)$. By Definition 1, changing w alone therefore leaves the value returned by F_v unchanged.

Lemma 2 (Unique evaluation). *For every causal model $M = (G, F)$ and exogenous state $u \in \mathcal{U}$, there is exactly one world $x \in \mathcal{X}$ satisfying*

$$x_v = F_v(u, x) \quad (v \in V).$$

Proof. Order the variables as $v_1, \dots, v_n$ so that every parent comes before its children. Construct a world in that order. At step k , fill the unassigned coordinates arbitrarily and evaluate F_{v_k} . Its parents already have values, so Lemma 1 makes the result independent of those temporary choices. Use it as x_{v_k} . The completed world satisfies every equation.

If x and y both satisfy the equations, induction along the same order shows that they agree. Indeed, they agree at the parents of v_k , so Lemma 1 gives $F_{v_k}(u, x) = F_{v_k}(u, y)$, and hence $x_{v_k} = y_{v_k}$.

Definition 3 (Run). For a model $M = (G, F)$ and exogenous state $u \in \mathcal{U}$, define $\text{Run}(M, u)$ to be the unique world $x \in \mathcal{X}$ such that

$$x_v = F_v(u, x) \quad \text{for every } v \in V.$$

3 The do-operator

An intervention supplies values at some variables and leaves all others unspecified. We represent it by listing values only for the variables it targets.

Definition 4 (Intervention). An intervention ι is a type-respecting partial map on V . Its domain $T_\iota := \text{dom}(\iota) \subseteq V$ is the target set, and $\iota(v) \in \mathcal{X}_v$ for every $v \in T_\iota$. We write $\iota_v := \iota(v)$. The empty intervention is the empty map, written $\emptyset$. For $S \subseteq V$, $\iota|_S$ denotes the ordinary restriction of ι to S . When $T_\iota = \{A\}$ and $\iota_A = a$, we write the intervention as $\text{do}(A = a)$.

Definition 5 (Graph surgery). For a directed graph G on V and a target set $T \subseteq V$, $\text{Surg}(G, T)$ has the same vertices as G and deletes exactly the arrows entering vertices in T .

Graph surgery depends only on the target set and does not record the values imposed there. Those values enter through the following change to the mechanisms.

Definition 6 (Mechanism replacement). For a mechanism family F on V and intervention ι , define F^ι by

$$F_v^\iota(u, x) := \begin{cases} \iota_v, & v \in T_\iota, \\ F_v(u, x), & v \notin T_\iota. \end{cases}$$

Example 1 (A three-variable intervention). Let $V := \{A, B, C\}$, let every endogenous value set be $\mathbb{R}$, and let $\mathcal{U} := \mathbb{R}^2$. For $u = (u_A, u_B)$, define

$$F_A(u, x) := u_A, \quad F_B(u, x) := x_A + u_B, \quad F_C(u, x) := 2x_B.$$

Thus $\text{Graph}(F)$ is the chain

$$A \longrightarrow B \longrightarrow C.$$

If $\iota := \text{do}(B = 7)$, then F_B^ι is the constant 7, while the other two mechanisms are unchanged. Hence $\text{Graph}(F^\iota)$ consists only of $B \rightarrow C$, exactly the graph obtained by deleting the arrow entering B . The updated equations have the unique solution

$$(A, B, C) = (u_A, 7, 14).$$

Theorem 1 (Graph–mechanism correspondence). For every mechanism family F on V and intervention ι ,

$$\text{Graph}(F^\iota) = \text{Surg}(\text{Graph}(F), T_\iota).$$

Proof. Fix two variables $w, v \in V$. If v is a target, then F_v^ι is constant, so no arrow enters v in $\text{Graph}(F^\iota)$; graph surgery likewise deletes every arrow entering v . If v is not a target, then $F_v^\iota = F_v$, so $w \rightarrow v$ is an arrow of $\text{Graph}(F^\iota)$ exactly when it is an arrow of $\text{Graph}(F)$; graph surgery leaves all such arrows unchanged. The two graphs therefore have the same arrows.

The identity compares the graphical and functional descriptions at the level of dependencies. A separate question is whether surgery on a graph G supplied with the model gives that same graph.

Corollary 1 (Agreement with the supplied graph). *Let $M = (G, F)$ be a causal model and let ι be an intervention. Then F^ι is compatible with $\text{Surg}(G, T_\iota)$. Moreover,*

$$\text{Graph}(F^\iota) = \text{Surg}(G, T_\iota)$$

if and only if every arrow of G that is absent from $\text{Graph}(F)$ enters an intervention target. Consequently, the equality holds for every intervention if and only if M is exact.

Proof. Compatibility says that $\text{Graph}(F)$ is a subgraph of G . Applying the same surgery to both graphs preserves this relation, and Theorem 1 identifies the smaller graph with $\text{Graph}(F^\iota)$. This proves compatibility after intervention.

The two surgically modified graphs are equal exactly when no extra arrow survives. An arrow survives surgery exactly when it does not enter a target. Thus equality holds exactly when every arrow of G absent from $\text{Graph}(F)$ enters a target. The uniform statement follows: exact models have no extra arrows, while equality for the empty intervention gives $\text{Graph}(F) = G$.

Example 2 (Why the supplied graph may differ). Let $V := \{A, B\}$, let the only arrow of G be $A \rightarrow B$, and suppose neither mechanism depends on an endogenous variable. Then $\text{Graph}(F)$ has no arrows, so (G, F) is compatible but not exact. An intervention that does not target B leaves $A \rightarrow B$ in $\text{Surg}(G, T_\iota)$, whereas $\text{Graph}(F^\iota)$ has no arrows. If the intervention targets B , both graphs have no arrows and therefore agree. Thus exactness is sufficient for agreement under every intervention, but it is not necessary for agreement under a particular one.

Having related the two separate edits, we now package their results as a complete intervened model.

Definition 7 (Do-operation). *For a causal model $M = (G, F)$ and intervention ι , define*

$$\text{Do}(M, \iota) := (\text{Surg}(G, T_\iota), F^\iota).$$

This pair is a causal model: deleting arrows preserves acyclicity, and Corollary 1 supplies compatibility.

The do-operation applies one edit to each component of the model. An alternative construction first replaces the mechanisms and then equips them with their exact dependency graph, giving $(\text{Graph}(F^\iota), F^\iota)$.

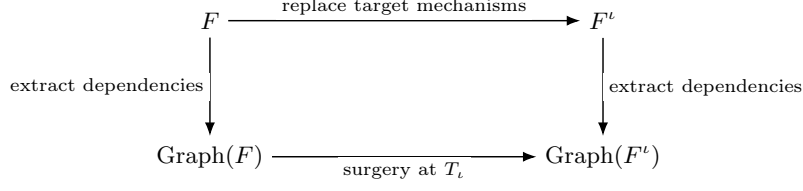


Fig. 1. The correspondence of Theorem 1. Replacing target mechanisms and extracting dependencies yields the same graph as extracting dependencies first and then performing surgery.

Corollary 2 (Agreement of the two constructions). *For every causal model $M = (G, F)$ and intervention ι , the pair $(\text{Graph}(F^\iota), F^\iota)$ is an exact causal model. For every exogenous state u ,*

$$\text{Run}(\text{Do}(M, \iota), u) = \text{Run}((\text{Graph}(F^\iota), F^\iota), u).$$

If M is exact, the stronger model equality also holds:

$$\text{Do}(M, \iota) = (\text{Graph}(F^\iota), F^\iota).$$

Proof. Corollary 1 says that $\text{Graph}(F^\iota)$ is a subgraph of the acyclic graph $\text{Surg}(G, T_\iota)$, so it is acyclic. The displayed pair is therefore a causal model, and it is exact by construction. Both models use the mechanism family F^ι , which determines their runs. If M is exact, Corollary 1 also gives $\text{Surg}(G, T_\iota) = \text{Graph}(F^\iota)$, so the graphs, and hence the models, are equal.

The intervened outcome is characterized directly by the following equations.

Corollary 3 (Intervention equation). *For every model $M = (G, F)$, intervention ι , exogenous state $u \in \mathcal{U}$, and world $x \in \mathcal{X}$,*

$$x = \text{Run}(\text{Do}(M, \iota), u) \iff \begin{cases} x_v = \iota_v, & v \in T_\iota, \\ x_v = F_v(u, x), & v \notin T_\iota. \end{cases}$$

Proof. By Definition 3, x is the run exactly when $x_v = F_v^\iota(u, x)$ for every v . At a target, this says $x_v = \iota_v$; at every other variable, it says $x_v = F_v(u, x)$. These are precisely the two displayed conditions.

The corollary gives the complete test for an intervened outcome: each target has its assigned value, and every equation outside the target set remains unchanged. Because G is acyclic, these conditions determine exactly one world.

4 Sequential interventions

Suppose ι is applied first and κ second. If both assign a value to the same variable, the later value from κ must prevail.

Theorem 2 (Sequential interventions). *For interventions ι and κ , define λ to be the partial map with domain $T_\iota \cup T_\kappa$ that agrees with κ wherever κ is defined and otherwise with ι . Then, for every model M ,*

$$\text{Do}(\text{Do}(M, \iota), \kappa) = \text{Do}(M, \lambda).$$

Proof. On both sides, the graph is obtained from G by deleting the arrows entering every variable targeted by either intervention. At a target of κ , the mechanism is the constant κ_v . At a target of ι but not κ , the constant ι_v remains. Every other mechanism is F_v . Thus both the graphs and the mechanism families are equal.

Corollary 4 (Basic intervention laws). *For every model M and interventions ι, κ :*

1. *The empty intervention does nothing: $\text{Do}(M, \emptyset) = M$.*
2. *Repeating an intervention changes nothing: $\text{Do}(\text{Do}(M, \iota), \iota) = \text{Do}(M, \iota)$.*
3. *Interventions with no common target may be applied in either order:*

$$\text{Do}(\text{Do}(M, \iota), \kappa) = \text{Do}(\text{Do}(M, \kappa), \iota).$$

4. *If both interventions target v , the later value wins: the mechanism at v in $\text{Do}(\text{Do}(M, \iota), \kappa)$ is the constant κ_v .*

Proof. The first statement follows directly from Definition 7, since surgery with an empty target set leaves G unchanged and $F^\emptyset = F$. The remaining statements follow from Theorem 2 by checking the value retained at each target.

For three or more interventions, the placement of parentheses does not matter: applying any finite sequence gives the same model as one intervention that collects all targets and keeps the last value assigned to each one. This is equality of the resulting models, not merely of their runs.

5 Dependence on ancestors

For an outcome set $O \subseteq V$, write $\text{An}_F(O)$ for the variables in O and every variable from which a directed path in the actual dependency graph $\text{Graph}(F)$ reaches a variable in O . If $v \in \text{An}_F(O)$, then every parent of v in $\text{Graph}(F)$ also lies in $\text{An}_F(O)$. Consequently, the equation for any variable in this set depends only on variables in the same set.

Theorem 3 (Dependence on ancestors). *Let $M = (G, F)$ be a causal model and let $O \subseteq V$. If two interventions ι and κ agree on $\text{An}_F(O)$, so that $\iota|_{\text{An}_F(O)} = \kappa|_{\text{An}_F(O)}$, then for every $u \in \mathcal{U}$,*

$$\text{Run}(\text{Do}(M, \iota), u)_O = \text{Run}(\text{Do}(M, \kappa), u)_O.$$

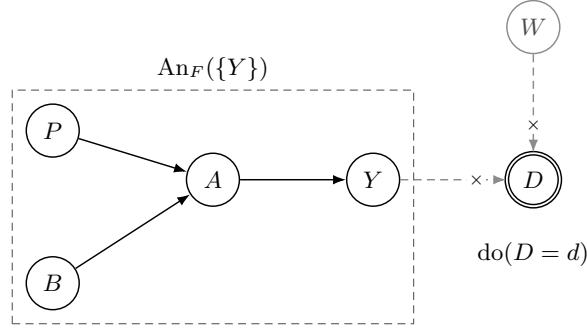


Fig. 2. The outcome at Y is determined inside its ancestor set. The dashed, crossed arrows are removed by $do(D = d)$; intervening on the downstream variable D cannot change Y .

Proof. Order the variables topologically in $Graph(F)$. We show by induction along this order that the two runs agree at every $v \in An_F(O)$.

If v is targeted by either intervention, then the assumption says it is targeted by both with the same value. The two runs agree at v . Otherwise both runs use the original mechanism F_v . By the definition of $An_F(O)$, every parent of v in $Graph(F)$ lies in $An_F(O)$, and all such parents precede v . The induction hypothesis gives equal values at these parents. Since $(Graph(F), F)$ is an exact causal model, Lemma 1 applied to that model gives equal values at v . Thus the runs agree throughout $An_F(O)$, and in particular on $O \subseteq An_F(O)$.

Corollary 5 (Discarding targets outside the ancestors). *For every model $M = (G, F)$, outcome set $O \subseteq V$, intervention ι , and exogenous state $u \in \mathcal{U}$,*

$$Run(Do(M, \iota), u)_O = Run(Do(M, \iota|_{An_F(O)}), u)_O.$$

In particular, if ι has no target in $An_F(O)$, then

$$Run(Do(M, \iota), u)_O = Run(M, u)_O.$$

Proof. The interventions ι and $\iota|_{An_F(O)}$ agree on $An_F(O)$. Apply Theorem 3. If ι has no target in $An_F(O)$, then $\iota|_{An_F(O)} = \emptyset$, and the second equality follows from the first law in Corollary 4.

Thus, to determine the outcome on O , one may discard all intervention targets outside $An_F(O)$.

The condition is sufficient, not necessary. Cancellation can make interventions on an actual ancestor irrelevant to an outcome. Take $V = \{A, B, Y\}$ and $\mathcal{U} = \mathcal{X}_A = \mathcal{X}_B = \mathcal{X}_Y = \{0, 1\}$, and define $F_A(u, x) = u$, $F_B(u, x) = x_A$, and $F_Y(u, x) = x_A \oplus x_B$. The model $(Graph(F), F)$ is exact and A is an ancestor of Y , but under $do(A = a)$ we have $B = a$ and hence $Y = a \oplus a = 0$. Thus $do(A = 0)$ and $do(A = 1)$ disagree on an ancestor while producing the same value of Y .

6 Related work

The do-operator is central to Pearl’s account of structural causal models [4]. In the usual structural-equation account, an intervention replaces selected equations by assigned values; deleting their incoming arrows is the corresponding graphical operation. Peters, Janzing, and Schölkopf give a modern treatment of functional models, interventions, and causal graphs [5], while Halpern gives axioms for reasoning with interventions in recursive models [1].

String-diagram accounts give explicit graphical languages for causal models. Jacobs, Kissinger, and Zanasi make the syntax–semantics separation formal: string diagrams are interpreted as stochastic matrices, and intervention is an operation on the diagrammatic syntax [2]. Lorenz and Tull develop the approach for a broader class of causal models [3]. Our result addresses a different, narrower question for ordinary deterministic structural causal models: how extensional dependencies change under constant mechanism replacement, including when a graph supplied with the mechanisms contains unused arrows.

Recent work uses mechanism functions to explain d -separation semantically [7]. Zhang’s earlier thesis describes incoming-edge deletion as the syntactic view of an intervention, constant replacement as its semantic view, and conjectures that the two are equivalent, with a formal proof in Coq proposed as the next step [6, Sec. 6.4]. Theorem 1 gives the conjecture a dependency-level formulation suitable for mechanization.

The same formulation distinguishes the dependency graph from a compatible supplied graph, which may contain unused arrows. Corollary 1 characterizes exactly when surgery on the two graphs agrees.

7 Conclusion

Graph surgery and constant mechanism replacement agree after dependency extraction:

$$\text{Graph}(F^\iota) = \text{Surg}(\text{Graph}(F), T_i).$$

For a graph supplied with the model, the analogous equality holds exactly when surgery removes all of its unused arrows; it therefore holds for every intervention if and only if the starting model is exact. The construction based on the supplied graph and the one based on the exact dependency graph always have the same run; for an exact starting model, they are the same intervened model.

The intervention equation characterizes the resulting outcome. Sequential interventions combine by retaining the last value assigned to each target, and an outcome depends only on interventions at its actual dependency ancestors. Together these results specify the do-operator directly while keeping its structural and functional components distinct.

Acknowledgements. This work was supported by the Anusandhan National Research Foundation (ANRF), Government of India, under the Prime Minister Early Career Research Grant ANRF/ECRG/2025/001136/ENS. I am grateful to Aalok Thakkar for helpful discussions.

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